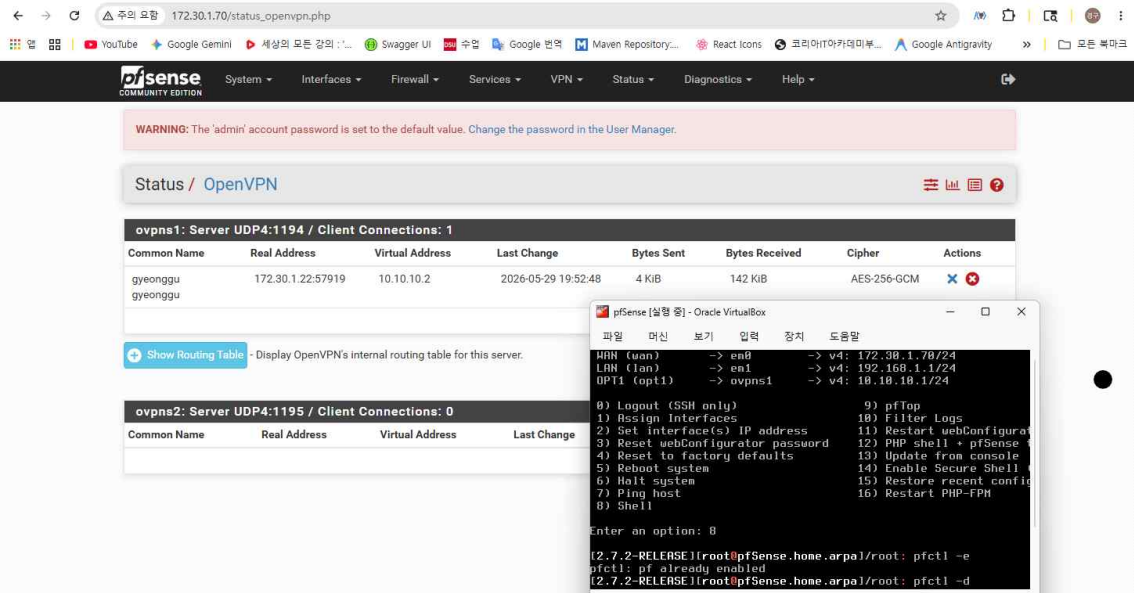
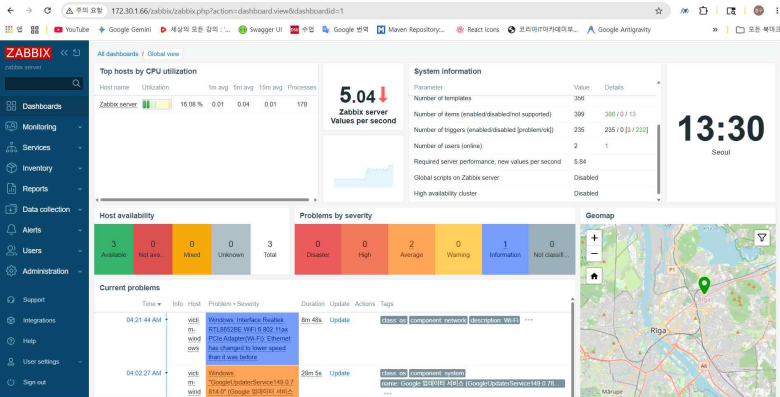
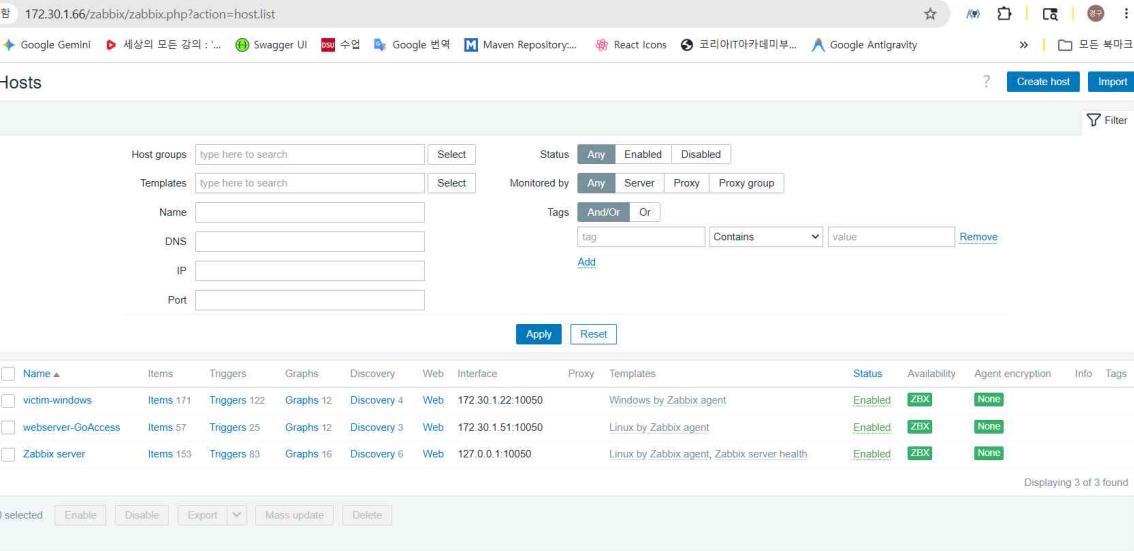


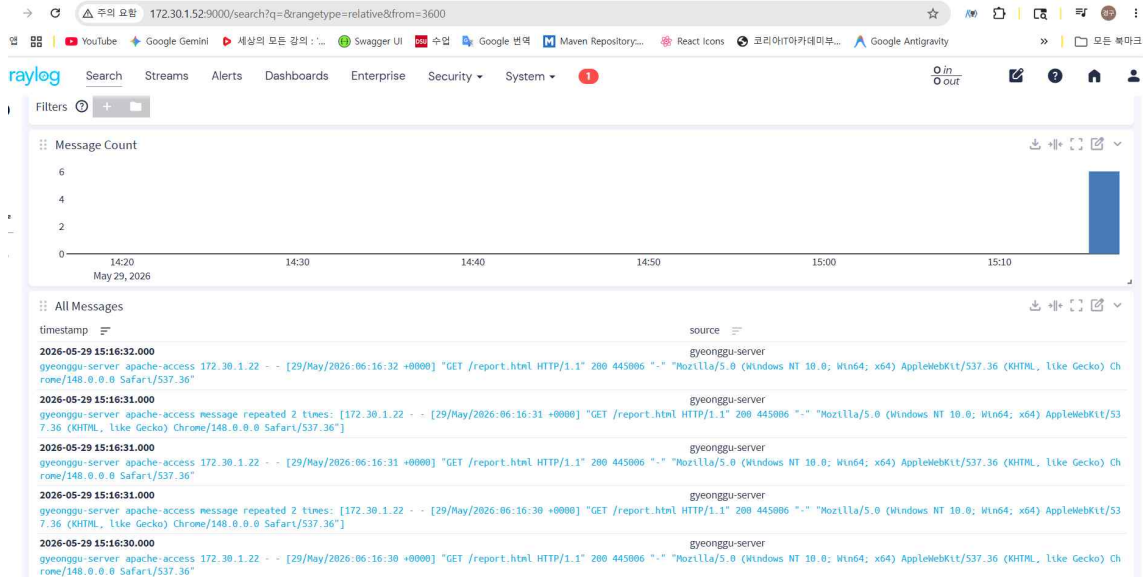
pfSense - OpenVPN 클라이언트 접속



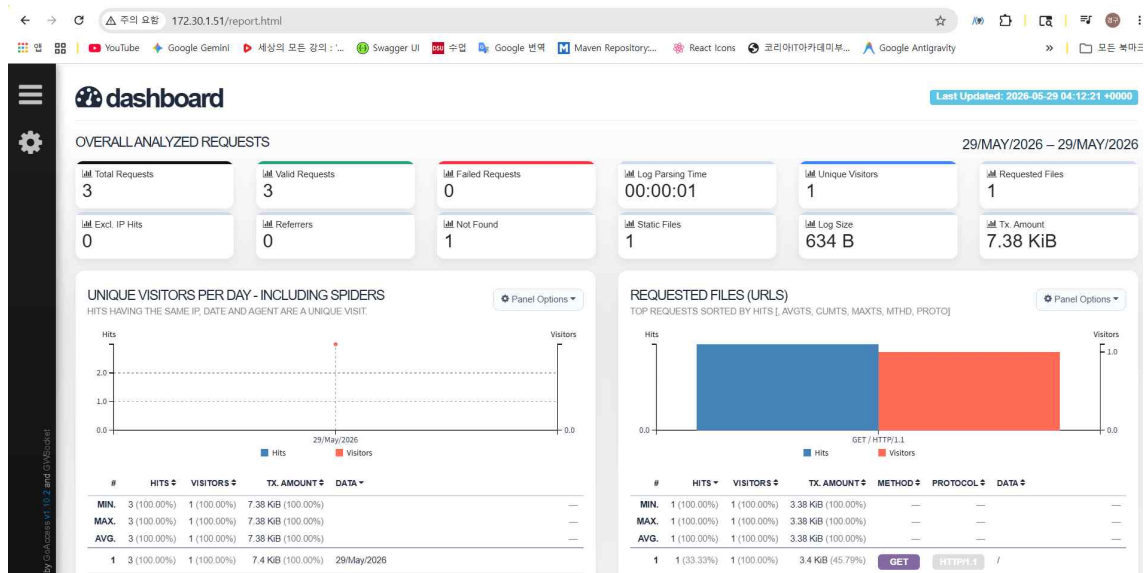
NMS-Zabbix Server 구축 후 GoAccess Agent 등록



SIEM-Wazuh 구축 후 Agent 등록 시 10점(용량 문제로 불가능할 경우 Graylog 구축)
용량이 부족하여 Graylog로 구축 후 goaccess 연동 로그 확인



WebServer-GoAccess 구축 후 웹 로그 분석 가능



Firewall-cmd rich-rule : Kali -> WebServer http 접속 불가능 정책 설정

```
zsh: corrupt history file /home/red/.zsh_history
(red@kali-Attacker)-[~]
$ sudo curl -I http://172.30.1.51
[sudo] password for red:
curl: (7) Failed to connect to 172.30.1.51 port 80 after 3 ms: Could not connect to server

(red@kali-Attacker)-[~]
$ sudo ping 172.30.1.51
PING 172.30.1.51 (172.30.1.51) 56(84) bytes of data:
64 bytes from 172.30.1.51: icmp_seq=1 ttl=64 time=0.475 ms
64 bytes from 172.30.1.51: icmp_seq=2 ttl=64 time=0.643 ms
64 bytes from 172.30.1.51: icmp_seq=3 ttl=64 time=0.728 ms
64 bytes from 172.30.1.51: icmp_seq=4 ttl=64 time=0.890 ms
64 bytes from 172.30.1.51: icmp_seq=5 ttl=64 time=0.763 ms
64 bytes from 172.30.1.51: icmp_seq=6 ttl=64 time=1.08 ms
64 bytes from 172.30.1.51: icmp_seq=7 ttl=64 time=0.716 ms
64 bytes from 172.30.1.51: icmp_seq=8 ttl=64 time=0.880 ms
64 bytes from 172.30.1.51: icmp_seq=9 ttl=64 time=0.816 ms
64 bytes from 172.30.1.51: icmp_seq=10 ttl=64 time=0.730 ms
^C
  -- 172.30.1.51 ping statistics --
  10 packets transmitted, 10 received, 0% packet loss, time 9106ms
 rtt min/avg/max/mdev = 0.475/0.771/1.077/0.152 ms

gyeonggu@gyeonggu-server:~$ sudo firewall-cmd --permanent --add-rich-rule='rule
family="ipv4" source address="172.30.1.35" port port="80" protocol="tcp" reject
success
gyeonggu@gyeonggu-server:~$ sudo firewall-cmd --reload
success
gyeonggu@gyeonggu-server:~$ sudo firewall-cmd --list-rich-rules
rule family="ipv4" source address="172.30.1.35" port port="80" protocol="tcp" r
ject
gyeonggu@gyeonggu-server:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:86:d5:3b brd ff:ff:ff:ff:ff:ff
    inet 172.30.1.61/24 brd 172.30.1.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe86:d53b/64 scope link
        valid_lft forever preferred_lft forever
gyeonggu@gyeonggu-server:~$
```

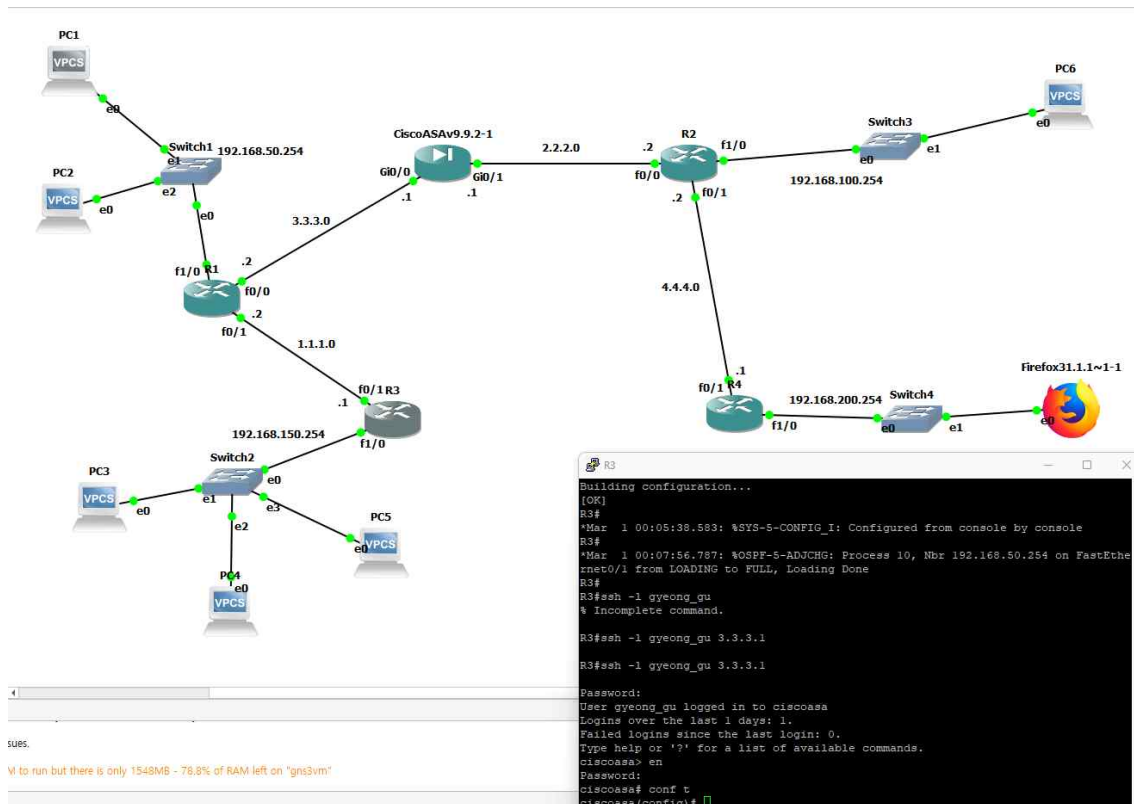
IPS 구축 및 Rule 설정

```
drop icmp any any -> 172.30.1.22 any (msg:"DDoS Smurf Attack Blocked"; detectio
n_filter:track by_src, count 20, seconds 5; sid:1000001; rev:2;)

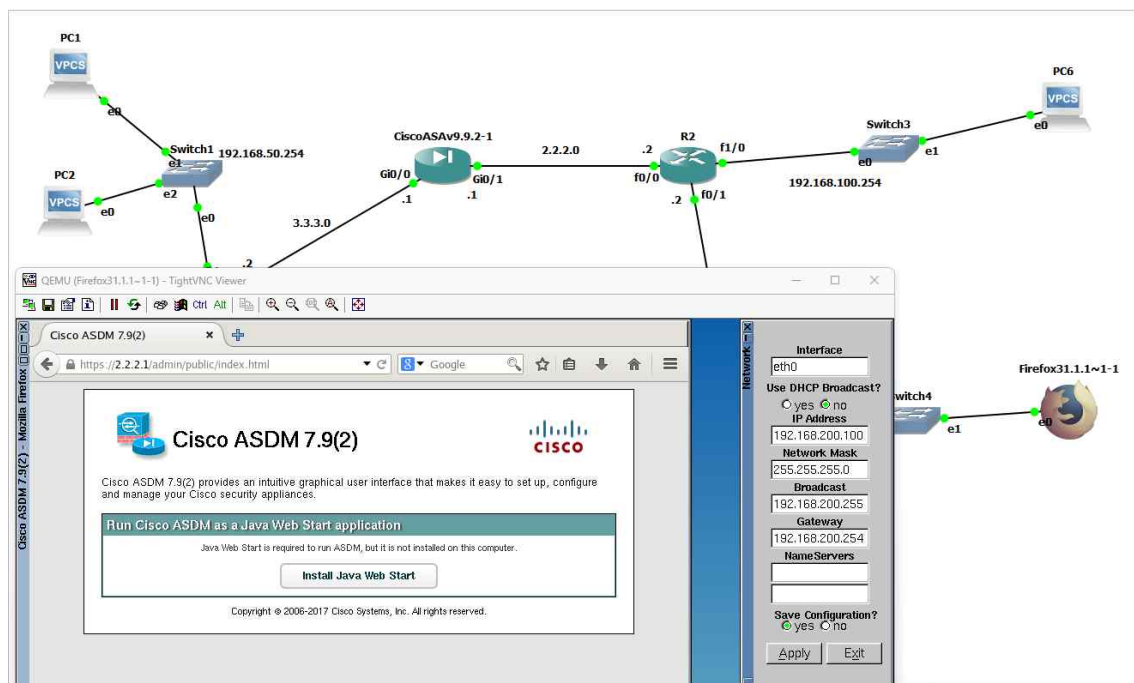
drop tcp 172.30.1.35 any -> 172.30.1.51 any (msg:"NMap Stealth SYN Scan Blocked"
; flags:S; detection_filter:track by_src, count 10, seconds 5; sid:1000002; rev:
2;)

drop tcp 172.30.1.35 any -> 172.30.1.66 80 (msg:"Zabbix Web Access Blocked"; co
ntent:"GET /zabbix"; nocase; sid:1000003; rev:2;)
```

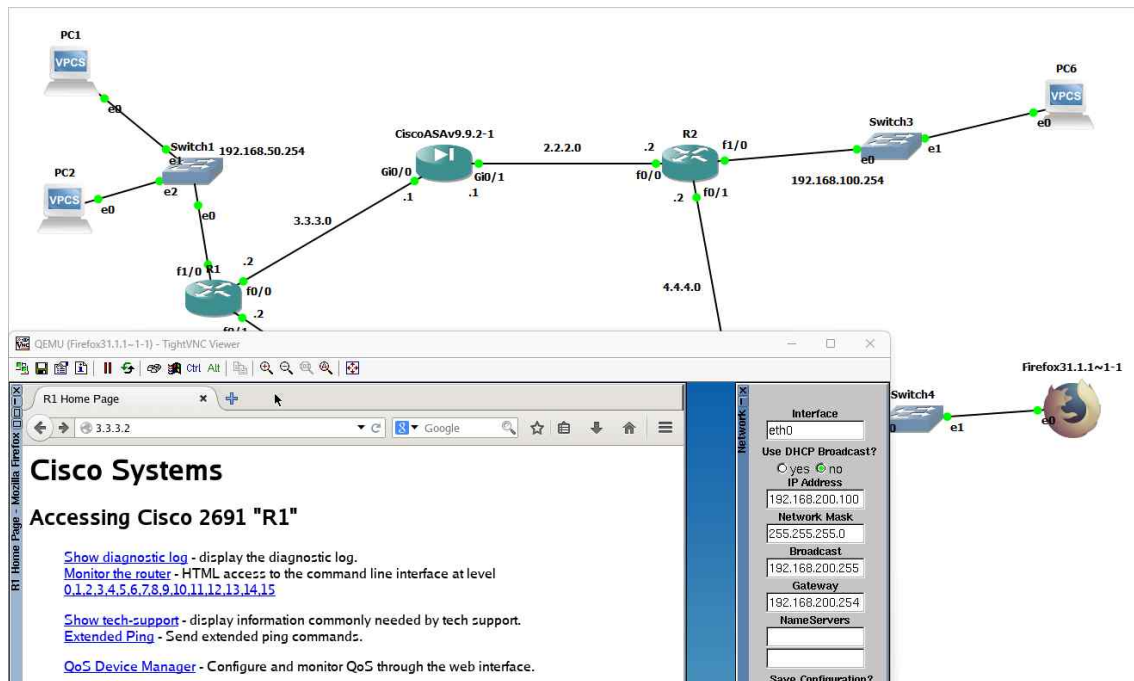
GNS3 서버 R3 -> inside ssh 접속



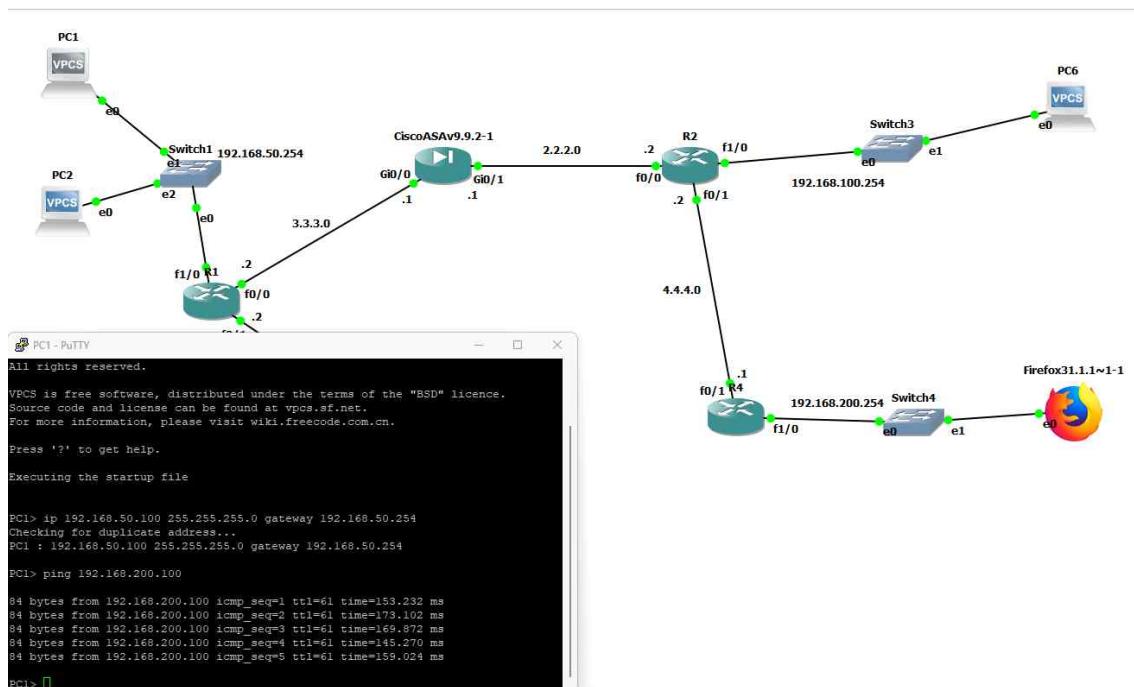
GNS3 서버 Firefox -> outside https 접속



GNS 서버 firefox -> R1 http 접속



GNS3 서버 PC1 -> firefox icmp 가능



Kali - Windows DDos Smurf 공격 IPS 확인

```
[red@kali-Attacker]~$ sudo hping3 -i -a 172.30.1.22 255.255.255.0 --fast
[sudo] password for red:
HPING 255.255.255.0 (eth0 255.255.255.0): icmp mode set, 28 headers + 0 data bytes
P[[A^][B^C
--- 255.255.255.0 hping statistic ---
1694 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

[red@kali-Attacker]~$ sudo hping3 -i -a 172.30.1.22 172.30.1.255 --fast
HPING 172.30.1.255 (eth0 172.30.1.255): icmp mode set, 28 headers + 0 data bytes
P[[A^][B^C
--- 172.30.1.255 hping statistic ---
52 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

[red@kali-Attacker]~$
```

```
05/29-07:27:15.881906 [Drop] [**] [1:1000001:2] DDoS Smurf Attack Blocked [**]
[PRIORITY: 0] (ICMP) 172.30.1.254 -> 172.30.1.22
05/29-07:27:15.982984 [Drop] [**] [1:1000001:2] DDoS Smurf Attack Blocked [**]
[PRIORITY: 0] (ICMP) 172.30.1.254 -> 172.30.1.22
05/29-07:27:16.082905 [Drop] [**] [1:1000001:2] DDoS Smurf Attack Blocked [**]
[PRIORITY: 0] (ICMP) 172.30.1.254 -> 172.30.1.22
05/29-07:27:16.185951 [Drop] [**] [1:1000001:2] DDoS Smurf Attack Blocked [**]
[PRIORITY: 0] (ICMP) 172.30.1.254 -> 172.30.1.22
05/29-07:27:16.287298 [Drop] [**] [1:1000001:2] DDoS Smurf Attack Blocked [**]
[PRIORITY: 0] (ICMP) 172.30.1.254 -> 172.30.1.22
05/29-07:27:16.394194 [Drop] [**] [1:1000001:2] DDoS Smurf Attack Blocked [**]
[PRIORITY: 0] (ICMP) 172.30.1.254 -> 172.30.1.22
05/29-07:27:16.497238 [Drop] [**] [1:1000001:2] DDoS Smurf Attack Blocked [**]
[PRIORITY: 0] (ICMP) 172.30.1.254 -> 172.30.1.22
05/29-07:27:16.595010 [Drop] [**] [1:1000001:2] DDoS Smurf Attack Blocked [**]
[PRIORITY: 0] (ICMP) 172.30.1.254 -> 172.30.1.22
05/29-07:27:16.694012 [Drop] [**] [1:1000001:2] DDoS Smurf Attack Blocked [**]
[PRIORITY: 0] (ICMP) 172.30.1.254 -> 172.30.1.22
05/29-07:27:16.796060 [Drop] [**] [1:1000001:2] DDoS Smurf Attack Blocked [**]
```

Kali - WebServer NMap Scanning IPS 확인

```
[red@kali-Attacker]~$ sudo nmap -sS -p- 172.30.1.51
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-29 03:29 -0400

[red@kali-Attacker]~$
```

```
05/29-07:29:50.524705 [Drop] [**] [1:1000002:2] NMap Stealth SYN Scan Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:55081 -> 172.30.1.51:61781
05/29-07:29:50.525001 [Drop] [**] [1:1000002:2] NMap Stealth SYN Scan Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:55081 -> 172.30.1.51:30577
05/29-07:29:50.525002 [Drop] [**] [1:1000002:2] NMap Stealth SYN Scan Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:55081 -> 172.30.1.51:37972
05/29-07:29:50.525233 [Drop] [**] [1:1000002:2] NMap Stealth SYN Scan Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:55081 -> 172.30.1.51:56875
05/29-07:29:50.525233 [Drop] [**] [1:1000002:2] NMap Stealth SYN Scan Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:55081 -> 172.30.1.51:37252
05/29-07:29:50.525477 [Drop] [**] [1:1000002:2] NMap Stealth SYN Scan Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:55081 -> 172.30.1.51:64985
05/29-07:29:50.525738 [Drop] [**] [1:1000002:2] NMap Stealth SYN Scan Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:55081 -> 172.30.1.51:38240
```

Kali - Zabbix Web 접속 IPS확인

```
Problem loading page x +
http://172.30.1.66/zabbix/
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

The connection was reset

The connection to the server was reset while the page was loading.
• The site could be temporarily unavailable or too busy. Try again in a few moments.
```

```
Preprocessor Object: SP REPUTATION Version 1.1 <Build 1>
Commenting packet processing (pid=2310)
Decoding Ethernet
05/29-07:39:30.044345 [Drop] [**] [1:1000022:2] Zabbix Web Access Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:43358 -> 172.30.1.66:80
05/29-07:39:30.047890 [Drop] [**] [1:1000022:2] Zabbix Web Access Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:43400 -> 172.30.1.66:80
05/29-07:39:30.052573 [Drop] [**] [1:1000022:2] Zabbix Web Access Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:43412 -> 172.30.1.66:80
05/29-07:39:30.056136 [Drop] [**] [1:1000022:2] Zabbix Web Access Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:43424 -> 172.30.1.66:80
05/29-07:39:30.061131 [Drop] [**] [1:1000022:2] Zabbix Web Access Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:43428 -> 172.30.1.66:80
05/29-07:39:30.067100 [Drop] [**] [1:1000022:2] Zabbix Web Access Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:43438 -> 172.30.1.66:80
05/29-07:39:30.072345 [Drop] [**] [1:1000022:2] Zabbix Web Access Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:43444 -> 172.30.1.66:80
05/29-07:39:30.077737 [Drop] [**] [1:1000022:2] Zabbix Web Access Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:43454 -> 172.30.1.66:80
05/29-07:39:30.081390 [Drop] [**] [1:1000022:2] Zabbix Web Access Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:43464 -> 172.30.1.66:80
05/29-07:39:30.086641 [Drop] [**] [1:1000022:2] Zabbix Web Access Blocked [**]
[PRIORITY: 0] (TCP) 172.30.1.35:43470 -> 172.30.1.66:80
```